

Pharmacoeconomics has been defined as the description and analysis of the cost of drug therapy to healthcare systems and society

Four Types of Pharmacoeconomic Analyses

Methodology	Cost Measurement Unit	Outcome Measurement Unit
Cost-minimization analysis(CMA)	Dollars	Assumed to be equivalent in comparable groups
Cost-benefit analysis(CBA)	Dollars	Dollars
Cost-effectiveness analysis(CEA)	Dollars	Natural units (life years gained, mm Hg blood pressure, mmol/L blood glucose)
Cost-utility analysis(CUA)	Dollars	Quality-adjusted life year(QALY) or other utilities

Direct medical costs

Are the costs incurred for medical products and services used to prevent, detect, and/or treat a disease.

Medications

Supplies

Laboratory tests

Healthcare professionals' time

Hospitalization

Direct medical costs

FIXED COSTS

"Overhead" costs
(e.g., heat, rent, electricity)

VARIABLE COST

Medications, fees for professional services, & supplies medications,

Direct nonmedical costs

Are any costs for nonmedical services that are results of illness or disease but do not involve purchasing medical services.

- **Transportation**
- **Food**
- **Family care**
- **Home aides**

Indirect nonmedical costs

Are the costs of reduced productivity (e.g., morbidity and mortality costs).

- ➡ **Lost wages (morbidity)**
- ➡ **Income forgone because of premature death (mortality)**

Intangible costs

Are those of other **non financial** outcomes of disease and medical care.

- ➡ **Pain**
- ➡ **Suffering**
- ➡ **Inconvenience**
- ➡ **Grief**

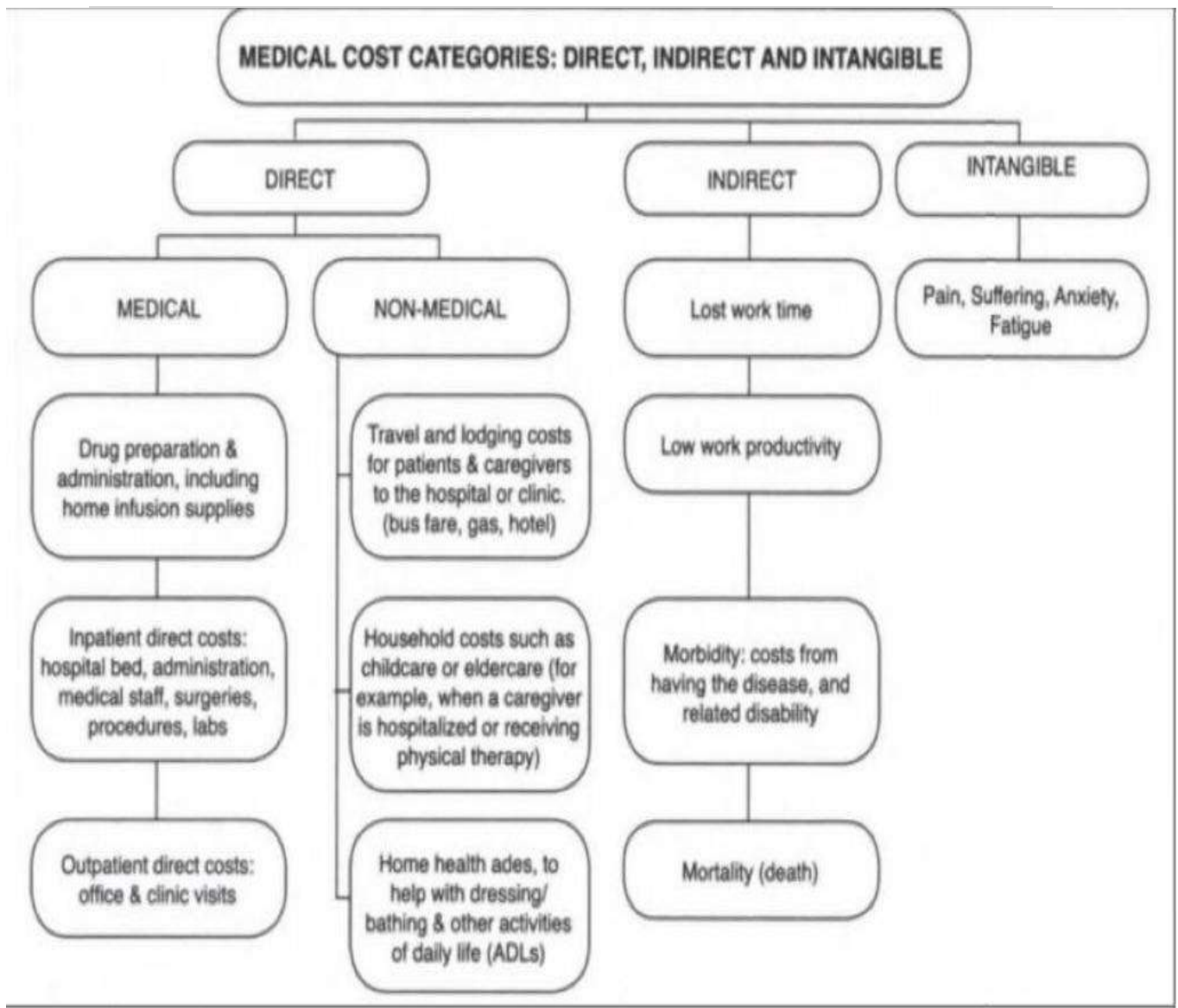
Opportunity costs

Represent the economic benefit forgone when using one therapy instead of the next best alternative therapy.

- ➡ **Lost opportunity**
- ➡ **Revenue forgone**

Incremental costs

Represent the additional cost that a service or treatment alternative imposes over another compared with the additional effect, benefit, or outcome it provides.



Economic outcomes

Are the direct, indirect, and intangible costs compared with the consequences of medical treatment alternatives.

Clinical outcomes

Are the medical events that occur as a result of disease or treatment (e.g., safety and efficacy end points).

Humanistic outcomes

Are the consequences of disease or treatment on patient functional status or quality of life along several dimensions (e.g., physical function, social function, general health and well-being, and life satisfaction)

A positive outcome

Is a desired effect of a drug (**efficacy or effectiveness measure**), possibly manifested as cases cured, life-years gained, or improved health-related quality of life (HRQOL).

A negative outcome

Is an **undesired or adverse effect** of a drug, possibly manifested as a treatment failure, an adverse drug reaction (ADR), a drug toxicity, or even death.

Methodology	Cost Minimization Analysis (CMA)	Cost Benefit Analysis (CBA)	Cost Effectiveness Analysis (CEA)	Cost Utility Analysis (CUA)
Type of Outcome	Equivalent Outcomes	Economic Outcomes	Clinical Outcomes	Humanistic Outcomes
Outcome Measurement Unite	Assumed to be Equivalent in comparable groups analysis	Monitory outcome Dollars \$	Natural Unites (mmHg blood analysis, mmol/L Blood glucose, cases cured yes or no)	Quality-adjusted life year (QALY)
Cost Measurement Unite	Dollars \$	Dollars \$	Dollars \$	Dollars \$
NOTE	Determination of the least costly alternative, the two or more alternatives must be equivalent therapeutically). shows only a "cost savings" of one program or treatment over another.	Measured by: •Human capital [HC] or •Willingness-to-pay [WTP]	Both must have the same unites to be compared Calculated by: •Average Cost Effectiveness Ratio (ACER) • Incremental Cost Effectiveness Ratio (ICER) symptoms free days (SFDs), % healed	Outcomes that assess patient's functional status or quality of life. QALY= Utility * Expected length of life Calculated by: • Rating scales (RS) • Standard gamble (SG) • Time tradeoff (TTO).
Example of the result	When comparing drug A to Drug B, Drug A save's you 5\$ each.	The benefit-to-cost ratio for hiring a pharmacist = 1.25 (Better) The benefit-to-cost ratio for the automated system = 1.15 (Worse)	When comparing drug A to Drug B, you have to pay an additional 6\$ per additional gastro-intestinal symptom free day	A person living for 20 years with arthritis is equivalent to a person living with perfect health for 10 years.

<i>Type of Cost Category</i>	<i>Examples</i>
Direct medical costs	Medications Medication monitoring Medication administration Patient counseling and consultations Diagnostic tests Hospitalizations Clinic visits Emergency department visits Home medical visits Ambulance services Nursing services
Direct nonmedical costs	Travel costs to receive health care (bus, gas, taxi) Nonmedical assistance related to condition (e.g., Meals-on-Wheels, homemaking services) Hotel stays for patient or family for out-of-town care Child care services for children of patients
Indirect costs	lost productivity for patient lost productivity for unpaid caregiver (e.g., family member, neighbor, friend) lost productivity because of premature mortality
Intangible costs	Pain and suffering Fatigue Anxiety



Which of the following pharmacoeconomic studies is the simplest because the outcomes are assumed to be equivalent and only the costs of intervention are compared ?

- A) Cost-utility analysis
- B) Cost-benefit analysis
- C) Cost-minimization analysis**

Which Of the following studies represent cost-effectiveness analyses study ?

- A) Comparing costs for 2 therapeutically equivalent products
- B) Comparing costs and savings of a pharmacist serves versus
- C) Comparing patients achieving control as the outcome for 2 anti hyperglycemic drugs**
- D) Comparison of costs of 2 chemotherapy regime with the cost C using best supportive care

202) Which of the following pharmacoeconomic studies is the simple conducted?

- A) Cost-Utility Analysis
- B) Cost-Benefit Analysis
- C) Cost-Minimization Analysis**
- D) Cost-Effectiveness Analysis

What is the type of costs for traveling from Jeddah to Riyadh to health care from pharmacoeconomic perspective?

- A) Indirect
- B) Intangible
- C) Direct -non medical**
- D) Non-direct medical

What kind of outcomes can be determined by cost-minimization?

- A) Monetary units (e.g. dollars)
- B) Quality- adjusted life-years (QALYS)
- C) Natural units (e.g. Life-years saved)
- D) Comparing the cost of two identical products (e.g. generic brand)**

Which of the following statements provides the best described incremental cost-effectiveness ratio?

- A) The cost per extra benefit of a new strategy, independent
- B) The extra cost to obtain an extra benefit when switching alternative to another**
- C) The cost per quality-adjusted life year gained

A pharmacist working in inpatient pharmacy received a request pharmacy and therapeutic (P&T) committee to evaluate a included in e hospital formulary.

The request wants to have the value in a monetary value(dollar value)

Which of the following pharmacoeconomics tools will the pharmacy to answer the request?

A) Cost-utility analysis

B) Cost-benefit analysis

C)Cost-minimization analysis

D)Cost-effectiveness analysis

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